
Table of Contents

EAS 4300 HW #8 Question 2	1
Given Parameters	1
900-Row Table of rf and rc	2
Inlet Conditions	2
Turbofan Cycle for Each Row	2
Best Combination of rc and rf	3
Figure 1: Overall Efficiency Surface	4
Figure 2: Overall Efficiency Contours	4
Save the 900-Row Table to CSV	5

EAS 4300 HW #8 Question 2

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Use the model developed in problem 1 to further optimize rc and rf at the optimum bypass ratio determined in problem 1. Do this using a single table where these values are varied across the following ranges: $1.5 < rf < 2.2$, $20 < rc < 28$. Make a table with 900 rows. Vary rf over the range using the "repeat pattern" across 30 rows. Vary rc over the range using the "apply pattern" across 30 rows. From this table solution, you can make an x-y-z plot, with $\eta_o(rc, rf)$. For the conditions and efficiencies listed in problem 1, determine if a different combination of rc and rf (other than 24 and 2) result in a further increase in the overall efficiency.

```
clear; clc; close all;
```

Given Parameters

```
M_f      = 0.8;           % Flight Mach number
Ta       = 223.252;      % Ambient temperature at 10 km [K]
Pa       = 26500;        % Ambient pressure at 10 km [Pa]
R        = 0.287;        % Gas constant [kJ/(kg*K)]
T04_max  = 1500;         % Turbine inlet temperature [K]
dhc      = 43000;        % Fuel heat of combustion [kJ/kg]
f_stoic  = 0.06;         % Stoichiometric fuel-to-air ratio
beta     = 5.88;         % Optimum bypass ratio from Question 1
rb       = 0.97;         % Burner pressure loss factor
```

% Efficiencies

```
eta_d    = 0.94;         % Diffuser efficiency
eta_c    = 0.87;         % Compressor efficiency
eta_b    = 0.98;         % Burner efficiency
eta_t    = 0.85;         % Turbine efficiency
eta_f    = 0.92;         % Fan efficiency
eta_cn   = 0.97;         % Core nozzle efficiency
eta_fn   = 0.98;         % Fan nozzle efficiency
```

% Specific heat ratios

```
% k(1) for inlet, diffuser, compressor, fan; k(2) for burner, turbine, nozzle
k = [1.4, 1.35];
Cp = (k ./ (k - 1)) * R;   % Specific heats [kJ/(kg*K)]
```

900-Row Table of rf and rc

rf takes 30 values that repeat every 30 rows ("repeat pattern"); rc takes 30 values, each held for 30 rows ("apply pattern").

```
N      = 30;
rfVec = linspace(1.5, 2.2, N)';
rcVec = linspace(20, 28, N)';
rf     = repmat(rfVec, N, 1);           % 1.5 ... 2.2, 1.5 ... 2.2, ...
rc     = repelem(rcVec, N);             % 20 (x30), 20.28 (x30), ...
nRows = numel(rf);
```

Inlet Conditions

Same inlet and diffuser as Question 1 ($T_{02} = T_{0a}$, adiabatic):

$$T_{0a} = T_a \left(1 + \frac{\gamma_1 - 1}{2} M^2 \right), \quad T_{02s} = T_a + \eta_d (T_{0a} - T_a), \quad P_{02} = P_a \left(\frac{T_{02s}}{T_a} \right)^{\gamma_1/(\gamma_1 - 1)}$$

```
T0a = Ta * (1 + (k(1)-1)/2 * M_f^2);
T02 = T0a;
T02s = eta_d * (T0a - Ta) + Ta;
P02 = Pa * (T02s/Ta)^(k(1)/(k(1)-1));
U    = M_f * sqrt(k(1) * R * 1000 * Ta); % Freestream velocity [m/s]
```

Turbofan Cycle for Each Row

The cycle is the Question 1 model, evaluated at $\beta = 5.88$ for each (r_c, r_f) pair:

$$T_{03} = T_{02} + \frac{T_{02} r_c^{(\gamma_1 - 1)/\gamma_1} - T_{02}}{\eta_c}, \quad T_{07} = T_{02} + \frac{T_{02} r_f^{(\gamma_1 - 1)/\gamma_1} - T_{02}}{\eta_f}$$

$$f = \frac{c_{p2} T_{04} - c_{p1} T_{03}}{\eta_b \Delta h_c - c_{p2} T_{04}}, \quad (1 + f) c_{p2} (T_{04} - T_{05}) = c_{p1} (T_{03} - T_{02}) + \beta c_{p1} (T_{07} - T_{02})$$

$$P_{05} = r_b r_c P_{02} \left(\frac{T_{05s}}{T_{04}} \right)^{\gamma_2/(\gamma_2 - 1)}, \quad T_{05s} = T_{04} - \frac{T_{04} - T_{05}}{\eta_t}$$

Both nozzles expand ideally to P_a (see Question 1), and

$$I = (1 + f) u_{e,hot} - u + \beta (u_{e,cold} - u), \quad \eta_0 = \eta_{th} \eta_p = \frac{I u}{f \Delta h_c}$$

```
eta_0 = zeros(nRows, 1);
I      = zeros(nRows, 1);
TSFC   = zeros(nRows, 1);

for j = 1:nRows
    % Compressor
    T03 = (T02 * rc(j)^((k(1)-1)/k(1)) - T02)/eta_c + T02;
```

```

Wc = Cp(1) * (T03 - T02);
P04 = rb * rc(j) * P02;

% Burner: Cp(1)*T03 + f*eta_b*dhc = (1 + f)*Cp(2)*T04
Fb = (Cp(2)*T04_max - Cp(1)*T03) / (eta_b*dhc - Cp(2)*T04_max);
T04 = T04_max;
if Fb > f_stoic
    Fb = f_stoic;
    T04 = (Cp(1)*T03 + f_stoic*eta_b*dhc) / ((1 + f_stoic)*Cp(2));
end

% Fan (bypass)
T07 = (T02 * rf(j)^((k(1)-1)/k(1)) - T02)/eta_f + T02;
Wf = beta * Cp(1) * (T07 - T02);
P07 = rf(j) * P02;

% Turbine drives compressor and fan
T05 = T04 - (Wc + Wf) / ((1 + Fb)*Cp(2));
T05s = T04 - (T04 - T05)/eta_t;
P05 = P04 * (T05s/T04)^(k(2)/(k(2)-1));

% Core and bypass nozzles, expanded to ambient
T6 = T05 - eta_cn * (T05 - T05 / (P05/Pa)^((k(2)-1)/k(2)));
Me = sqrt((2*(T05/T6 - 1))/(k(2)-1));
T8 = T07 - eta_fn * (T07 - T07 / (P07/Pa)^((k(1)-1)/k(1)));
M8 = sqrt((2*(T07/T8 - 1))/(k(1)-1));
Ue_hot = Me * sqrt(k(2)*R*1000*T6);
Ue_cold = M8 * sqrt(k(1)*R*1000*T8);

% Performance
I(j) = (1 + Fb)*Ue_hot - U + beta*(Ue_cold - U);
TSFC(j) = Fb / I(j);
KE = (1 + Fb)*Ue_hot^2 + beta*Ue_cold^2 - (beta + 1)*U^2;
eta_th = KE / (2*Fb*dhc*1000);
eta_p = 2*I(j)*U / KE;
eta_0(j) = eta_th * eta_p;
end

```

Best Combination of rc and rf

```

[eta_0_max, iBest] = max(eta_0);
[~, iBase] = min(abs(rc - 24) + abs(rf - 2)); % grid point closest to rc =
24, rf = 2
fprintf('Maximum overall efficiency: eta_0 = %.4f at rc = %.2f, rf = %.3f\n',
...
    eta_0_max, rc(iBest), rf(iBest));
fprintf('Near the Question 1 design (rc = %.2f, rf = %.3f): eta_0 = %.4f\n',
...
    rc(iBase), rf(iBase), eta_0(iBase));

% Reshape the table into a 30 x 30 grid: rows follow rf, columns follow rc
ETA0 = reshape(eta_0, N, N);
[RC, RF] = meshgrid(rcVec, rfVec);

```

Maximum overall efficiency: $\eta_{a0} = 0.2867$ at $r_c = 28.00$, $r_f = 1.886$
Near the Question 1 design ($r_c = 23.86$, $r_f = 2.007$): $\eta_{a0} = 0.2808$

Figure 1: Overall Efficiency Surface

```
figure;
surf(RC, RF, ETA0);
xlabel('Compressor Pressure Ratio (r_c)');
ylabel('Fan Pressure Ratio (r_f)');
zlabel('\eta_0');
title('Overall Efficiency \eta_0(r_c, r_f) at \beta = 5.88');
shading interp;
colorbar;
grid on;
```

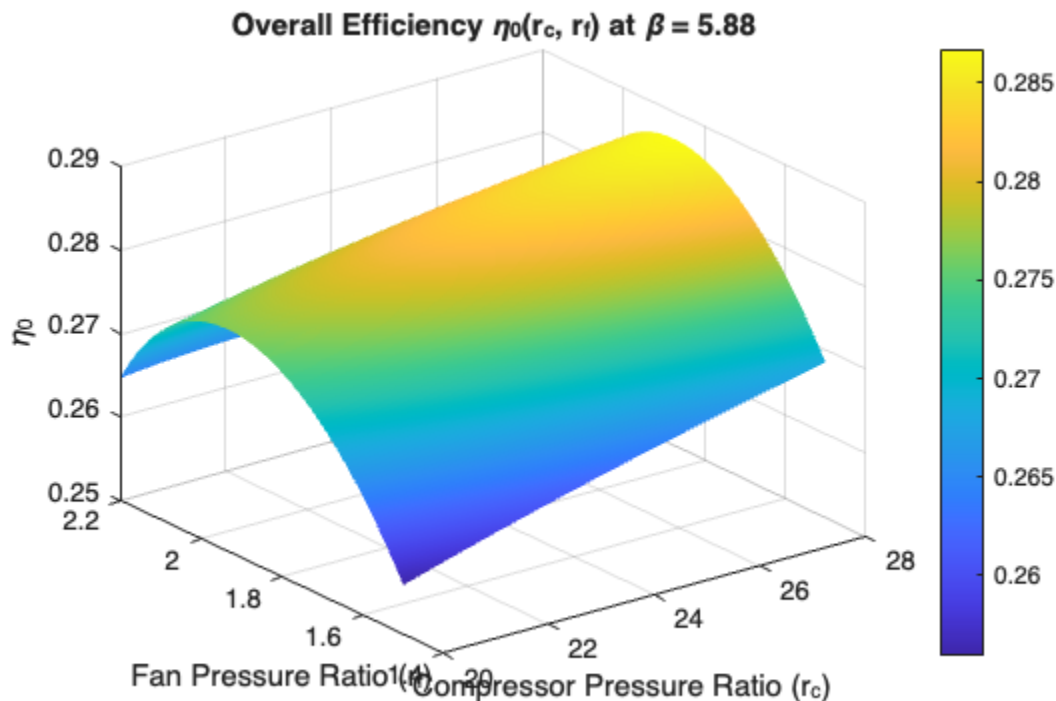


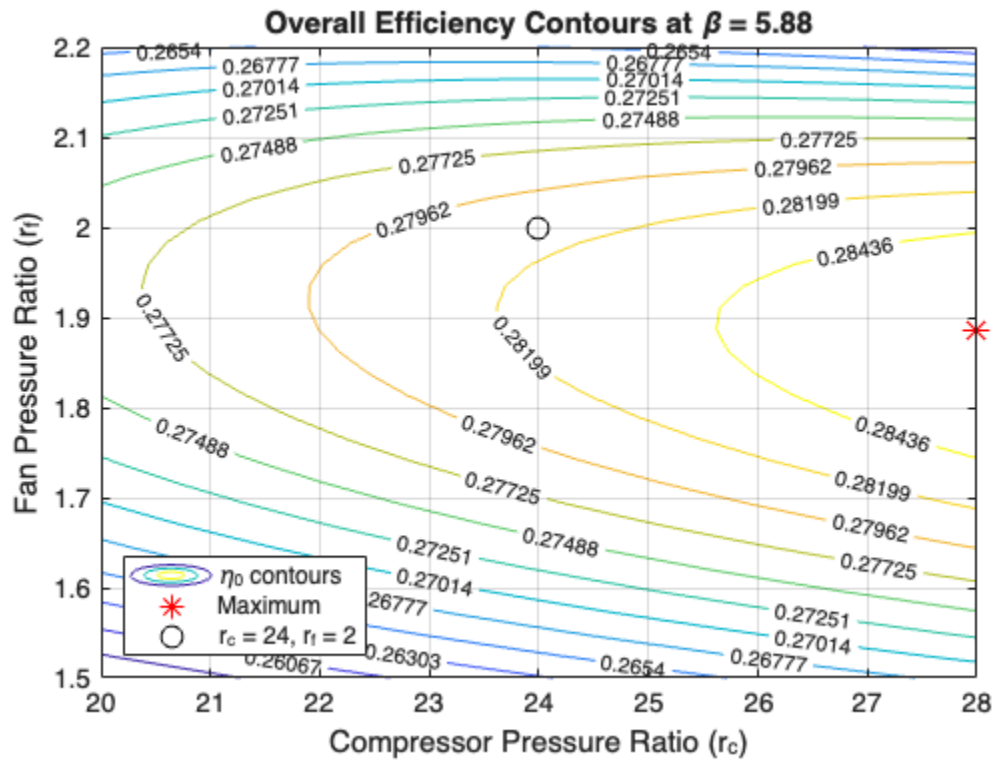
Figure 2: Overall Efficiency Contours

```
figure;
[C, h] = contour(RC, RF, ETA0, 12);
clabel(C, h, 'FontSize', 8);
hold on
plot(rc(iBest), rf(iBest), 'r*', 'MarkerSize', 10);
plot(24, 2, 'ko', 'MarkerSize', 8);
hold off
legend('\eta_0 contours', 'Maximum', 'r_c = 24, r_f = 2', 'Location',
'southwest');
xlabel('Compressor Pressure Ratio (r_c)');
```

```

ylabel('Fan Pressure Ratio (r_f)');
title('Overall Efficiency Contours at \beta = 5.88');
grid on;

```



Save the 900-Row Table to CSV

```

dataTable = table(rf, rc, eta_0, I, TSFC, ...
    'VariableNames', {'FanPressureRatio', 'CompPressureRatio',
    'OverallEfficiency', ...
    'SpecificThrust', 'TSFC'});
writetable(dataTable, 'HW8_Q2_data.csv');
disp('Data saved to HW8_Q2_data.csv');

```

Data saved to HW8_Q2_data.csv

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